LAB 8 : Server Configuration – DHCP, DNS and Web Server in Cisco Packet Tracer

Objectives:

- To configure a server to provide DHCP, DNS, and Web (HTTP) services in Cisco Packet Tracer
- To understand how DHCP assigns IP addresses and how DNS resolves domain names
- To verify the proper functioning of server services by accessing a web server from client devices

Theory:

Server Configuration:
Server configuration refers to preparing a dedicated system to deliver essential network services to client devices. To ensure consistent and dependable access, servers are typically assigned fixed IP addresses. In this experiment, a single server was set up to handle DHCP, DNS, and HTTP services. A properly configured server allows centralized administration, improves communication efficiency, and makes network management more straightforward.

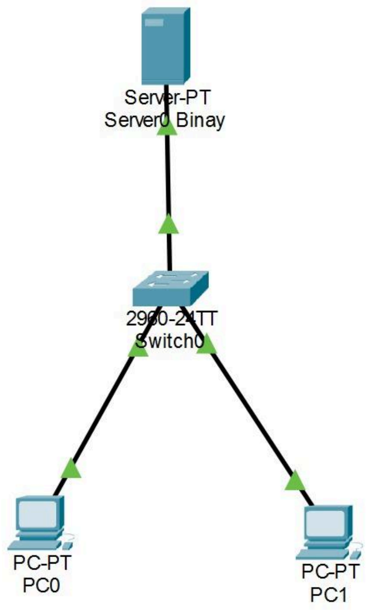
DHCP (Dynamic Host Configuration Protocol):
DHCP is used to automatically provide network settings to client devices, including IP address, subnet mask, default gateway, and DNS information. This automation removes the need for manual configuration, minimizes configuration mistakes, and supports easy network expansion. In this lab setup, the server dynamically assigned IP addresses to all connected client systems.

DNS (Domain Name System):
The Domain Name System enables users to access network resources using easy-to-remember names instead of numerical IP addresses. The DNS server maintains records that associate domain names with their corresponding IP addresses and responds to client requests by providing the correct address. In this experiment, DNS allowed clients to access the server using a defined domain name.

Web Server (HTTP Service):
A web server is responsible for storing and delivering web pages to users through the HTTP protocol. When a client sends a request via a web browser, the server responds by providing the requested content, such as HTML files. In this lab, enabling the HTTP service allowed users to successfully view a webpage hosted on the server.

Integration of DHCP, DNS, and Web Services:
These network services function together to create a smooth and user-friendly network environment. DHCP handles automatic IP assignment, DNS manages name resolution, and the web server delivers requested content. Their combined operation demonstrates how centralized services enhance network efficiency, reduce configuration effort, and improve overall usability.

Network Topology :



Configuration :

IP Addressing Scheme

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

For PCs,

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

For Server,

Device	Default Gateway	DNS Server
Server0	192.168.1.1	192.168.1.2

Procedure :

Server Configuration :
Step 1 : Assign Static IP to Server

1. Click on Server
2. Go to Desktop → IP Configuration
3. Configure the following:

   - IP Address: 192.168.1.2
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1
   - DNS Server: 192.168.1.2

DHCP Configuration :
Step 2 : Enable DHCP Service

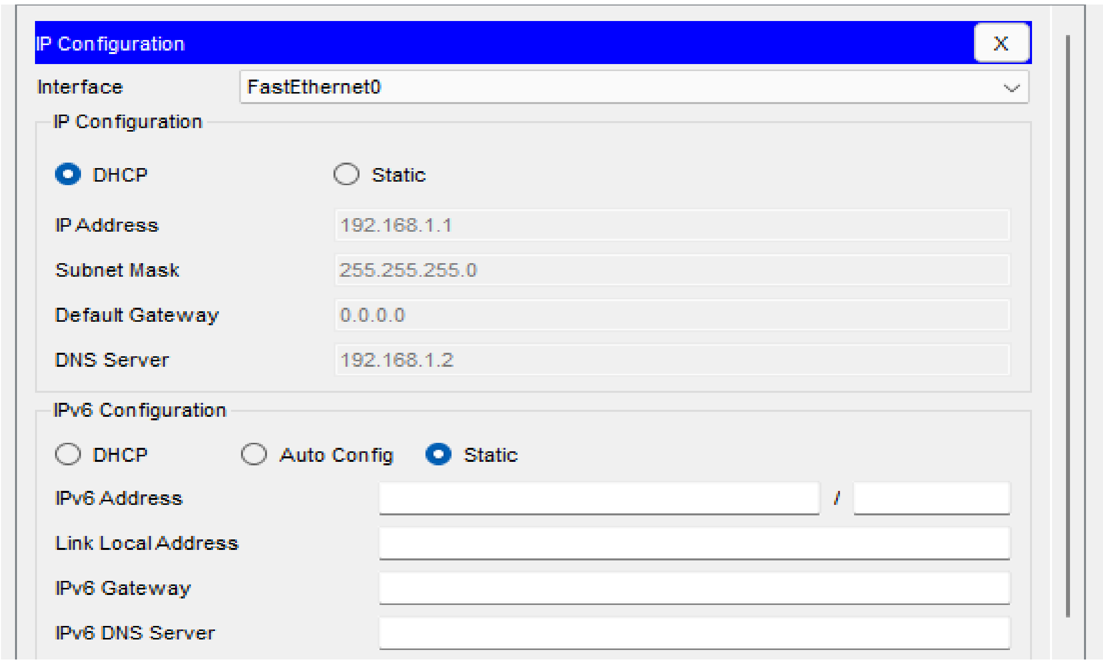
1. Go to Server → Services → DHCP
2. Turn DHCP: ON

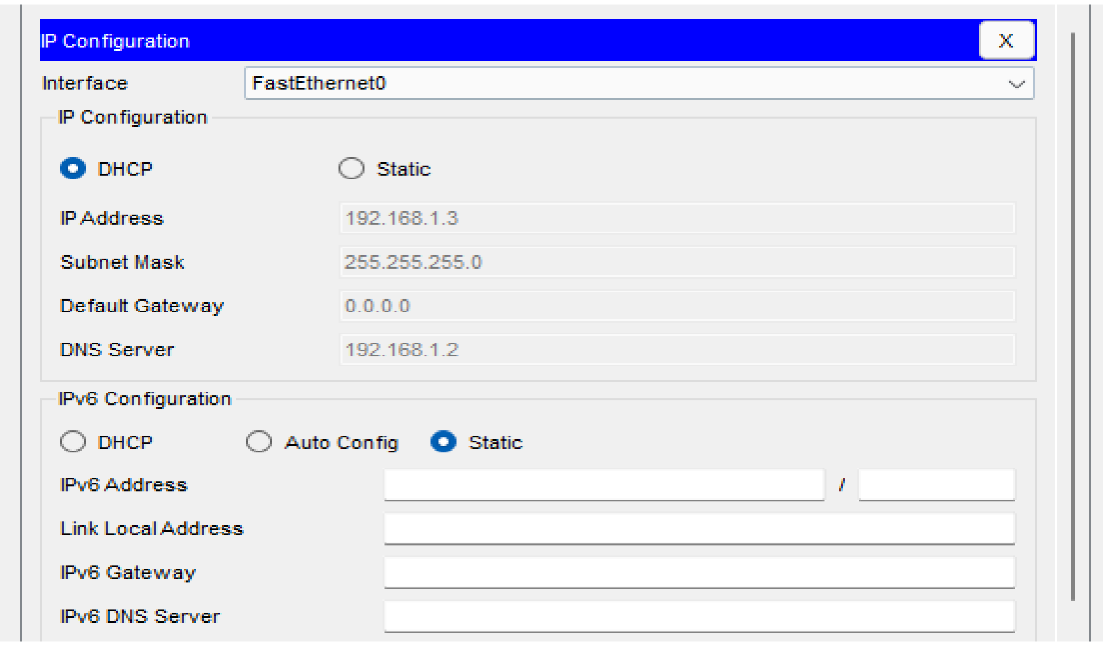
Step 3 : Create DHCP Pool

Field	Value
Pool Name	LAB_POOL
Default Gateway	192.168.1.1
DNS Server	192.168.1.2
Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add

Verification :
On each PC:
- Go to Desktop → IP Configuration
- Select DHCP






```
C:\>ping www.binay.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

DNS Server Configuration :
Step 4 : Enable DNS Service

1. Go to Server → Services → DNS
2. Turn DNS: ON

Step 5 : Add DNS Records

- Name: www.hcoe.com
- Type: A record
- Address: 192.168.1.2

Click Add

Verification :
On each PC:

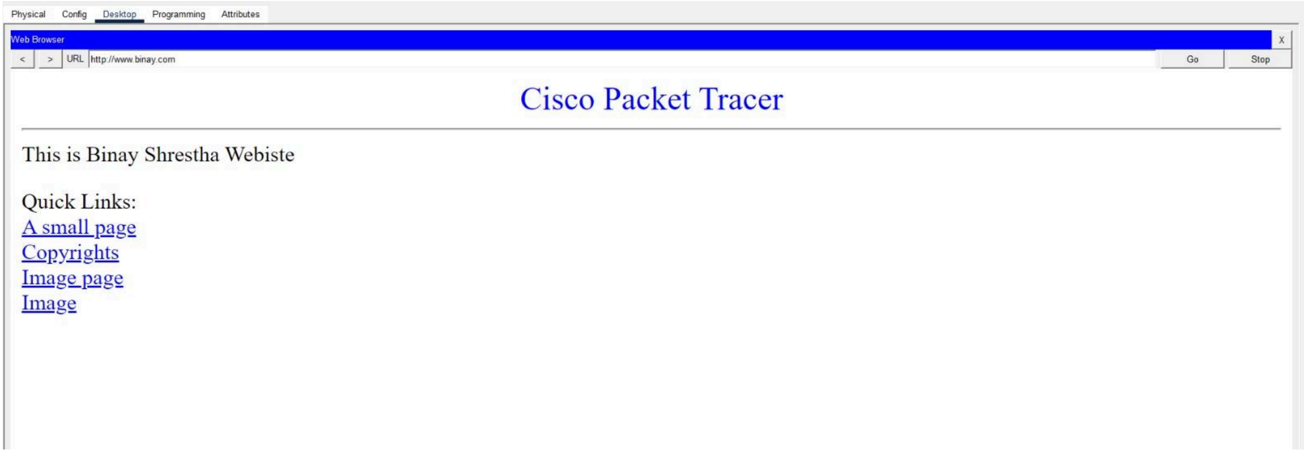
- Open Command Prompt
- Ping: www.hcoe.com

```
C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Web Server Configuration :
Step 6 : Enable HTTP Service

1. Go to Server → Services → HTTP
2. Turn HTTP: ON
3. Edit index.html (optional)

Verification :
On PC:
- Go to Desktop → Browser
- Type: http://www.binay.com



RESULT :

The server was properly set up to provide DHCP, DNS, and web services. Client computers obtained their IP addresses automatically through the DHCP service. The DNS service successfully translated the domain name into the correct server IP address, and the website loaded without issues using HTTP, confirming that all configured services were functioning correctly.

Discussion:

This laboratory exercise focused on setting up and testing core network services and observing how they work together within a network. DHCP reduced manual configuration by automatically assigning IP addresses, while DNS allowed devices to access resources using readable domain names. The web server setup confirmed successful communication at the application layer. Overall, the lab helped develop a clearer understanding of centralized network services and introduced basic methods for identifying and resolving network issues in a simulated setup.

Conclusion:

The configuration and testing of DHCP, DNS, and HTTP services in Cisco Packet Tracer were completed successfully. The results confirmed proper IP address distribution, accurate name resolution, and reliable access to hosted web content. This experiment highlighted the role of integrated network services in ensuring efficient and functional modern network infrastructures.